Optimization

Calculus $\longrightarrow$ Optimization.

Functions

$y = f(x)$

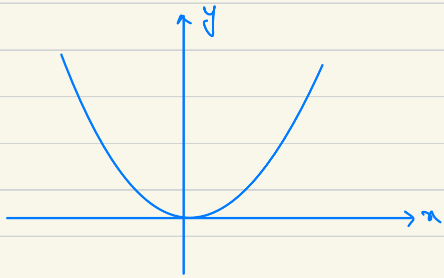
input $\longrightarrow$ $\boxed{\text{rule}}$ $\longrightarrow$ Output
(x) f(x) (y).

Single variable.

$y = f(x) = x^2 + 2x$

$f(2) = (2)^2 + (2)(2)$

$f(2) = 8$



- Given same i/p gives same o/p.

Single Variable $f(x)$

Multiple Variable $z = f(x,y)$

$z = f(x,y) = x^2 + y^2$

Graph $\rightarrow$ 3d- surface graph

- Given a set of i/p we get some o/p.

$$f(x_1, x_2, \text{------} x_n)$$

n+1 dims graph.

Parameters in functions :

- Variables used to determine the behaviour of fxn.

$$y = f(x) = 5\,x^2$$

variable.

parameter

$$y = f(x) = ax^2$$

ML Models as Mathematical Function :

ML models are basically mathematical fxns.

cgpa	iq	package.
9	90	5
8	100	8
10	120	16

placement = f (cgpa , iq)

Parametric and Non-Parametric Model.

Functions

Parametric model	Non-parametric model.
→ Data distribution assumption.	→ No assumptions.
→ Fixed no. of parameters irrespective of no. of rows in data.	→ Not fixed parameters.
(eg. Linear Regression)	(eg. Decision Tree)

cgpa	iq	Placement
8	80	

<u>Model</u>

↓

linear regression.

placement = f (cgpa, iq)

↑

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

↑

$\boxed{\beta_0, \beta_1, \beta_2}$ ⇒ Parameters.

Linear Regression as a Parametric ML Model :

cgpa	iq	placement (y)	$\hat{y}$
–	–	–	–
–	–	–	–
–	–	–	–

$\beta_0, \beta_1, \beta_2$

1.) Start with some random value. (y)

2.) Calculate the prediction ($\hat{y}$)

$(\beta_0, \beta_1, \beta_2)$

B ———— A

Loss fxn.

Loss function :

A loss fn also known as cost fn or objective fn, is a mathematical fn that measures the difference b/w the $\hat{y}$ and y in ML model. The main goal of training a ML model is to minimize a value of the loss fn, which corresponds to improve the model's performance on the given task.

Loss fn. play a crucial role in the optimization process, guiding the learning algo. to adjust the model's parameters to achieve better predictions.

y	$\hat{y}$
8	11
9	6
6	3

A Loss fn may be :

$y-\hat{y}$ or $|y-\hat{y}|$ or

$(y-\hat{y})^2$ or $\sqrt{(y-\hat{y})^2}$

Optimization simply means to find a fn that gives least diff. b/w the actual & predicted $\longrightarrow y$ & $\hat{y}$

There are many loss fns. in ML exists.

$$\Sigma |y-\hat{y}|$$

minimum.

How to select a good loss function :

Calculating Parameters from a Loss Function (the easy way and problem)

cgpa	iq	placement	$\hat{y}$
8	80	8	11
7	70	7	6
6	60	6	3

$\beta_0, \beta_1, \beta_2 = 1, 2, 3$ (let)

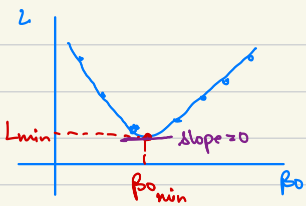
let loss fxn:

$$\sum (y - \hat{y})^2$$

$$E = \frac{3^2 + 1^2 + 3^2}{3} = 6.3.$$

$$\boxed{L(\beta_0, \beta_1, \beta_2) = \underset{\beta_0\,\beta_1\,\beta_2}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

for i in range(1, n):
    Change $\beta_0, \beta_1, \beta_2$ values (say only β_0 to simplify)
    results = new predictions ($\hat{y}$)
    these $\hat{y}$ in $L(\beta_0, \beta_1, \beta_2)$ gives new value of errors.
Plot the graph β_0 and Loss.



← parabolic due to sq.

For finding Lmin, we do optimization mathematically.

$$\frac{dL}{\partial \beta_0} = 0 \quad \text{(finding slope)}$$

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\hat{y}_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Let.
$(\beta_0 = ?, \ \beta_1 = 1, \ \beta_2 = 2)$

$$\hat{y}_i = \beta_0 + x_1 + 2x_2$$

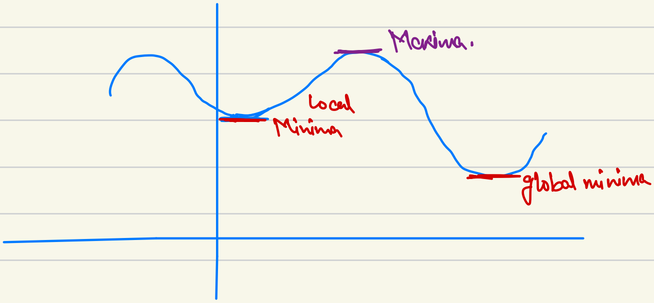
$$\mathcal{L} = \sum_{i=1}^{n}(y_i - \beta_0 - x_1 - 2x_2)^2$$

$$\frac{dL}{d\beta_0} = 2(y_i - \beta_0 - x_1 - 2x_2)\cdot(-1) = 0$$

$$\frac{dL}{d\beta_0} = -2(y_i - \beta_0 - x_1 - 2x_2) = 0$$

$$\beta_0 = y_i - x_1 - 2x_2$$

Is this a full proof approach?



We may stop at a sub-optimal result (like at local minima).

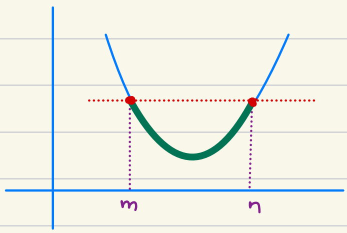
Problems with the easy way.

1.) Non- convexity. (might have many local minima & maxima)
2.) Complexity. (hard to differentiate)
3.) Scalability. (High dimensional feature space eg. big matrix)
4.) Online learning and streaming data.

Convex and Non- Convex Loss Functions :

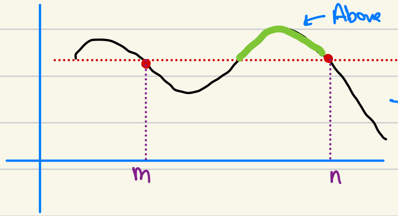
Convex fxn : At any time the green shaded protion should not surpass the red line between points (m , n). This property should hold true.

$L(\beta_0)$



→ Convex fxn
→ Only 1 minima.

← Above the red line b/w point (m,n)



→ Non convex fxn.
→ Multiple local minima.

Optimization Techniques :

1.) Gradient Descent. (Mostly used in ML and DL)
2.) Newton Method.
3.) Quadratic - programming.
4.) Linear - programing.

Gradient Descent :

- Linear Regression + its varients (Ridge, Lasso, Multiple etc.)
- Logistic Regression.
- Perceptoon
- Deep Learning (ANN, CNN, RNN, Advanced NN architectures)

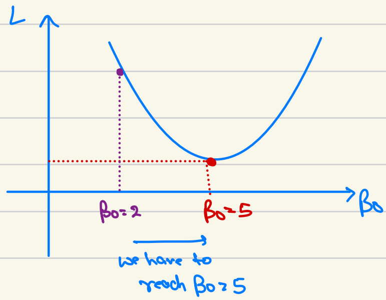
Quadratic Techniques :

- SVM
- PCA

Newton Methods :

- first Order Derivative → Gradient Descent.
- Dual differentiation

Gradient Descent (idea) :



Find the slope and move to opposite direction.

i.e. $\frac{dL}{d\beta}$.

$$\beta_{0_{new}} = \beta_{0_{old}} - \frac{dL}{d\beta}$$

Learning Rate :

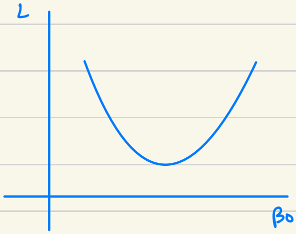
$$\beta_{0_{new}} = \beta_{0_{old}} - \underset{\text{Learning Rate.}}{\eta}\, \frac{dL}{d\beta_0}$$

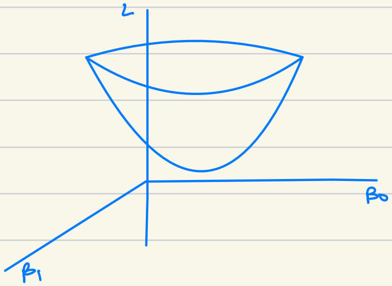
- We will always reach minima, never the maxima.
- We might not reach global minima.
  To fix this we use varients of gradient descent like
  Momentum, NAG, Adam, RMS prop. etc.

Gradient Descent with Multiple Parameters :

$L(\beta_0) \longrightarrow$ I/p cols $\rightarrow x_1, x_2, x_3 \ldots x_n$

Parameters $\beta_0, \beta_1, \ldots \ldots \beta_n \rightarrow (n+1)$ parameters.





Partial derivative of β_0, β_1

$$\frac{\partial L}{\partial \beta_0}, \quad \frac{\partial L}{\partial \beta_1}$$

$$\beta_0 = \beta_0 - \eta \frac{\partial L}{\partial \beta_0}$$

$$\beta_1 = \beta_1 - \eta \frac{\partial L}{\partial \beta_1}$$

$$\nabla L(\beta_0, \beta_1, \beta_2, \ldots \beta_n) = \left(\frac{\partial L}{\partial \beta_0}, \frac{\partial L}{\partial \beta_1}, \ldots \frac{\partial L}{\partial \beta_n} \right)$$

Problems faced in Optimization (gradient descent) :

1.) Non-convexity : → multiple local minima.
→ saddle point (m(slope) at point close to 0 | global minima)

2.) Ill-conditioning : Loss fxn (L) is moved only at one direction, when gradient in one direction is much larger than other.
Oscillate & converge slowly.

3.) Vanishing & exploding gradient : when m is either too low or high, we do not reach the optimize loss fxn.

4.) Overfitting.

5.) Scalability : In DL, in a neural network there are thousands of weights, updating each weight (i.e. calculating thousands of differentiations) computationally expensive.

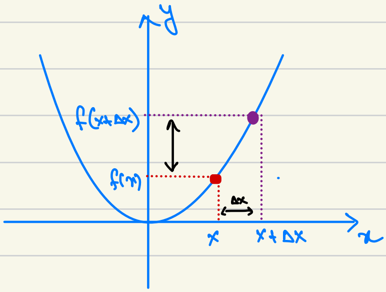
Constrained Optimization :
SVM | PCA.

Constrain like $\hat{y} = 2.$ ← condition should remain true.
$(y - \hat{y})$.

Differential Calculus

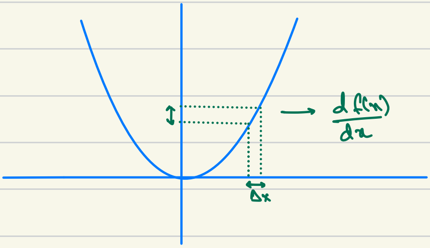
Derivative $\longrightarrow$ instantaneous rate of change.

$$y = f(x) = x^2$$



$$\text{rate of change} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$\text{instantaneous rate of change} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

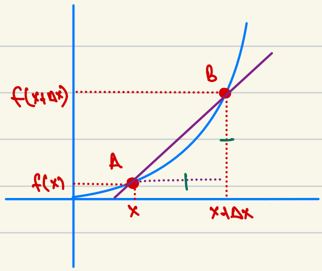


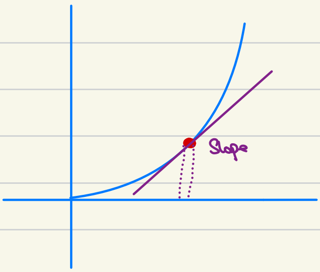
$\longrightarrow \dfrac{d\,f(x)}{dx}$

$\Delta x \to 0.1$

$dx \to 0.0000000\ldots01$

Slope (m)





$dx \to 0$

$$\text{slope} = \frac{f(x+\Delta x) - f(x)}{\Delta x} = \tan\theta$$

Slope of line

$$y = x^2 + 2x$$

$$\frac{dy}{dx} = 2x + 2 = 0 \text{ [slope]}$$

$$x = -1 \qquad \text{[minima]}$$

Q. 1.) $y = f(x) = x^2$

$$\frac{df(x)}{dx} = \frac{f(x+dx) - f(x)}{dx}$$

$$f(x+dx) = (x+dx)^2$$

$$f(x) = x^2$$

$$= \frac{(x+dx)^2 - x^2}{dx}$$

$$= \frac{x^2 + dx^2 + 2x\,dx - x^2}{dx}$$

$$= \frac{dx^2 + 2x\,dx}{dx}$$

$$= dx + 2x$$

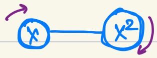
As dx is close to 0.

$$\therefore = 2x.$$

Final Intution :

$$y = f(x) = x^2$$

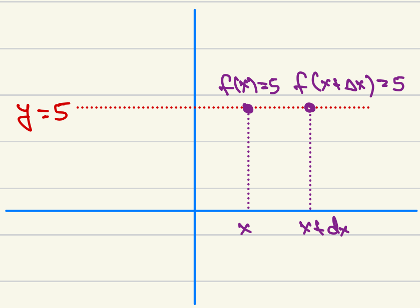
$$\frac{dy}{dx}$$



If we rotate x, how much rotation will be at x^2 knob and in which direction.

Derivative of a constant :

$$\frac{d(c)}{dx} = 0$$



$$\frac{dy}{dx} = \frac{f(x+\Delta x) - f(x)}{\Delta x} = \frac{5-5}{\Delta x} = 0$$

Common Derivatives :

$$\frac{d(x)}{dx} = 0$$

$$\frac{d}{dx}(\log_a(x)) = \frac{1}{x\ln(a)}$$

$$\frac{d(\sin x)}{dx} = \cos x$$

$$\frac{d(\cos x)}{dx} = -\sin x$$

$$\frac{d(\tan x)}{dx} = \sec^2 x$$

$$\frac{d(\sec x)}{dx} = \sec x \tan x$$

$$\frac{d(\csc x)}{dx} = -\operatorname{cosec} x \cot x$$

$$\frac{d(\cot x)}{dx} = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(a^x) = a^x \ln(a)$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}, \quad x > 0$$

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

Power Rule :

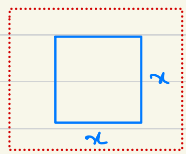
$$\frac{d}{dx}(x^n) = n x^{n-1}$$

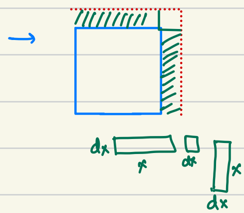
$$\frac{dx^2}{dx} = 2x$$

$$\frac{dx^3}{dx} = 3x^2$$

$$\frac{dx^6}{dx} = 6x^5$$

$f(x) = x^2$





Area = x^2 .

$$\frac{dy}{dx} = \frac{f(x+dx) - f(x)}{dx}$$

$$f(x+dx) = x^2 + xdx + dx^2 + xdx$$

$$= x^2 + 2xdx + \underline{\underline{dx^2}} \longrightarrow \text{very small}$$

$$= x^2 + 2xdx$$

$$\therefore \quad \frac{dy}{dx} = \frac{\cancel{x^2} + 2xdx - \cancel{x^2}}{dx}$$

$$\frac{dy}{dx} = \frac{2xdx}{dx}$$

$$\frac{dy}{dx} = 2x$$

Sum Rule :

$$(f(x) \pm g(x))' = f'(x) + g'(x)$$

$$\frac{d(x^2 + \log x)}{dx} = \frac{d(x^2)}{dx} + \frac{d \log(x)}{dx}$$

$$= 2x + \frac{1}{x}$$

Product Rule :

$$(f(x)\, g(x))' = f(x)'\, g(x) + f(x)\, g(x)'$$

$$(c f(x))' = c\,(f'(x))$$

$$\frac{dy}{dx} = x^2 \operatorname{Sin} x$$

$$\frac{d(x^2 \operatorname{Sin} x)}{dx} = \frac{d\, x^2}{dx} \cdot \operatorname{Sin} x + x^2 \frac{d}{dx}(\operatorname{Sin} x)$$

$$= 2x \operatorname{Sin} x + x^2 \operatorname{Cos} x$$

Quotient Rule :

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)\, g(x) - f(x)\, g'(x)}{[g(x)]^2}$$

$$\frac{d}{dx}\left(\frac{x^2}{\operatorname{Sin} x}\right) = \frac{\frac{d}{dx}(x^2) \cdot \operatorname{Sin} x - x^2 \frac{d}{dx} \operatorname{Sin} x}{\operatorname{Sin}^2 x}$$

$$= \frac{2x \operatorname{Sin} x - x^2 \operatorname{Cos} x}{(\operatorname{Sin} x)^2}$$

Chain Rule :

$$\frac{d}{dx}(f(g(x))) = f'(g(x))\, g'(x)$$
$$= \frac{df}{dg} \cdot \frac{dg}{dx}$$

① $$\frac{d\,(\sin(x^2))}{dx} = \frac{d \sin x^2}{d x^2} \cdot \frac{d x^2}{dx}$$
$$= \cos x^2 \cdot 2x$$
$$= 2x \cos x^2$$

② $$\frac{d(\sin \log x^2)}{dx} = \frac{d(\sin \log x^2)}{d \log x} \cdot \frac{d(\log x^2)}{dx^2} \cdot \frac{dx^2}{dx}$$

$$= \cos \log x^2 \cdot \frac{1}{x^2} \cdot 2x$$

$$= \frac{2 \cos \log x^2}{x}$$

- In DL → Used in backpropagation.

Partial Differentiation :

$y = f(x)$ $\qquad\qquad \dfrac{dy}{dx} = f'(x)$

Single variable

More than 1 variable.

$y = f(x_1, x_2)$

$y = f(x_1, x_2, x_3)$

$z = f(x,y) = x^2 + y^2$. [3d parabola].

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial y} = 2y$$

$$\boxed{\frac{\partial z}{\partial x} \quad . \quad \frac{\partial z}{\partial y}}$$

Partial derivative

Slope at $(1,2)$

$$\frac{\partial z}{\partial x} = 2x = 2(1) \Rightarrow 2.$$

$$\frac{\partial z}{\partial y} = 2y = 2(2) \Rightarrow 4.$$

Higher Order Derivative :

$y = x^3$

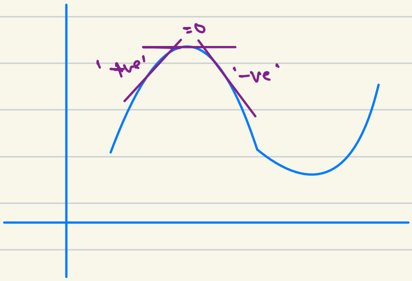
$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d\frac{(x^3)}{dx}}{dx} = \frac{d(3x^2)}{dx} = 3 \cdot 2x = 6x.$$

Instantaneous rate of change of $\underbrace{\text{slope}}_{\text{derivative.}}$.

Slope = 0 $\quad \frac{dy}{dx} \rightarrow$ maxima | minima.

$\frac{d^2y}{dx^2} \rightarrow$ maxima.



Matrix Differentiation :

$$\frac{d}{dx} Ax$$

Matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{array}{c} x \\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{array}$$

$$Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix} \begin{array}{l} \to f_1 \\ \to f_2 \end{array}$$

$$\begin{bmatrix} \frac{df_1}{dx_1} & \frac{df_1}{dx_2} \\ \frac{df_2}{dx_1} & \frac{df_2}{dx_2} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\boxed{\frac{d(Ax)}{dx} = A}$$

$$a = x^T A x$$

$$y = x^T A x$$

A is symmetric i.e $A^T = A$. eg. $\begin{bmatrix} A_{11} & A_x \\ A_x & A_{22} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_x \\ A_x & A_{22} \end{bmatrix}$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_{11} & a_x \\ a_x & a_{11} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \underset{1 \times 2}{\begin{bmatrix} x_1 & x_2 \end{bmatrix}} \underset{2 \times 1}{\begin{bmatrix} a_{11} x_1 + a_x x_2 \\ a_x x_1 + a_{11} x_2 \end{bmatrix}}$$

$$= a_{11} x_1^2 + a_x x_1 x_2 + a_x x_1 x_2 + a_{11} x_2^2$$

$$= a_{11} x_1^2 + 2 a_x x_1 x_2 + a_{11} x_2^2$$

$$= \begin{bmatrix} \frac{dy}{dx_1} \\ \frac{dy}{dx_2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{d(a_{11} x_1^2 + 2 a_x x_1 x_2 + a_{11} x_2^2)}{dx_1} \\ \frac{d(a_{11} x_1^2 + 2 a x_1 x_2 + a_{11} x_2^2)}{dx_2} \end{bmatrix}$$

$$= \begin{bmatrix} 2 a_{11} x_1 + 2 a_x x_2 \\ 2 a_{11} x_2 + 2 a_x x_1 \end{bmatrix}$$

$$= 2 \begin{bmatrix} a_{11} x_1 + a_x x_2 \\ a_{11} x_2 + a_x x_1 \end{bmatrix}$$

$$= 2 \begin{bmatrix} a_{11} & a_x \\ a_x & a_{11} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= (2 A x)^T$$

$$= 2 x^T A^T$$

(as $A^T = A$ for symmetric matrix)

$$\boxed{\frac{\partial \alpha}{\partial x} = 2 x^T A}$$